Data Mining

Outlier Analysis

Outlier Analysis

- Outlier: data point significantly deviating from the rest
- Applications:
  - Fraud detection
  - Fault diagnosis
  - Intrusion detection
  - Data cleaning
- Outliers may represent errors, rare events, or new phenomena

Types of Outliers

- Point Outliers: individual data instances deviating from others
- Contextual (Conditional) Outliers: abnormal only in specific context (e.g., time, location)
- Collective Outliers: a group of data instances behaving unusually together
- High-Dimensional Outliers: harder to detect due to sparse distances

Outlier Detection Algorithms

- Z-Score / Statistical: assumes known distribution
- k-NN Distance-Based: high average distance → outlier
- LOF (Local Outlier Factor): local density deviation
- Isolation Forest: isolates anomalies quickly using random splits
- One-Class SVM / Autoencoder: model-based detection for complex data

Outlier Detection Algorithms

- **Statistical methods:**
  - **Z-Score:** Calculates how many standard deviations a data point is from the mean. Points with a high Z-score are considered outliers.
  - **Probabilistic and Statistical Models:** Uses models like the Gaussian distribution to identify data points with a low probability of occurring.
  - **Interquartile Range (IQR):** Identifies outliers as points that fall below and above the IQR.

Outlier Detection Algorithms

- **Model-Based Methods:**
  - **Isolation Forest:** Randomly isolates data points. Outliers are easier to isolate and therefore require fewer splits in the random trees.
  - **One-Class Support Vector Machine (OC-SVM):** Creates a boundary around the normal data points, and any points outside this boundary are flagged as outliers.

Outlier Detection Algorithms

- **Proximity-Based Methods:**
  - **Distance-Based (e.g., KNN):** Identifies points that are far from their K nearest neighbors.
  - **Density-Based (e.g., DBSCAN):** Clusters data based on density and flags points that do not belong to any cluster as outliers.

Outlier Detection Algorithms

- **Visual Methods:**
  - **Scatter Plots:** Shows the relationship between two variables, with outliers appearing as points far from the main cluster.
  - **Box Plots:** Displays the distribution of a dataset, with individual points plotted outside the "whiskers" indicating potential outliers.
  - **Histograms:** Illustrates the frequency distribution of a variable, making it easy to see data points in low-frequency bins